Anonymous Supplement to the Second-Round Reviewer Response

This supplement contains the detailed numerical tables, threshold-sensitivity check, candidate-DAG schematic, and implementation-scope comparison referenced in the main OpenReview response.

Table S1. Dispersion for manuscript Table 2

Values are terminal risk ratios at budget two. Means and sample standard deviations are computed over 25 independently parameterized instances sharing one complete five-node CPDAG. The SD uses denominator 24; graph, outcome, and random-action uncertainty is integrated within each instance.

Policy	Exact label: mean +/- SD	10% label error: mean +/- SD
Lookahead attribution risk	0.02096 +/- 0.01420	0.15491 +/- 0.02259
One-step attribution risk	0.02304 +/- 0.01723	0.15529 +/- 0.02268
Posterior width	0.02269 +/- 0.01769	0.16297 +/- 0.02663
Graph information	0.02322 +/- 0.01720	0.17454 +/- 0.02607
Sign-pattern information	0.07157 +/- 0.06740	0.24911 +/- 0.08485
Random	0.13613 +/- 0.01989	0.28727 +/- 0.02271

The figure bands use the instance mean plus or minus 1.96 standard errors, namely mean +/- 1.96 SD/sqrt(25).

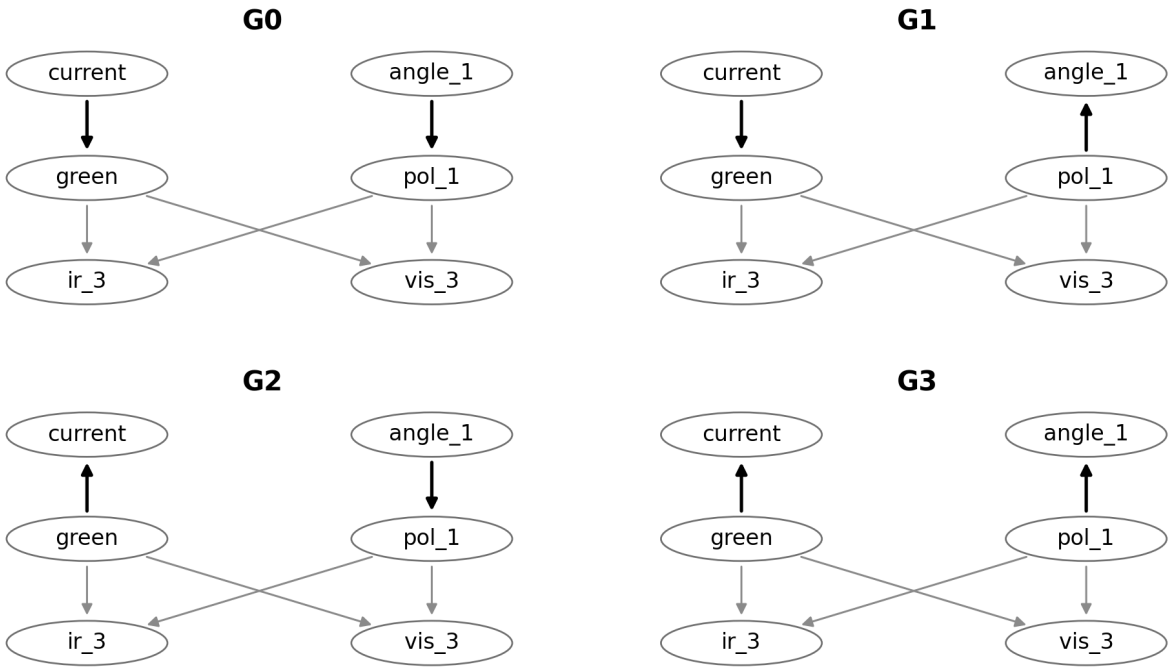
Table S2. Threshold sensitivity for measured responses

The screening rule is |mean(z_I)-mean(z_R)| > k SE, with SE = sqrt(s_I^2/n_I + s_R^2/n_R). Here n_I = 1,000 and n_R = 10,000. Multipliers k = 2, 3, 4, and 5 produce the same candidate-set trajectory, 4 -> 2 -> 1.

Intervention and response	Mean difference	Estimated SE	|Delta|/SE
green -> current	10.99850	0.23591	46.62117
pol_1 -> angle_1	58.22300	0.51832	112.32929

This check describes stability for the supplied observations and is used only to support the Causal Chamber filtering protocol.

Figure S1. Supplied Causal Chamber candidate DAGs



Gray arrows are shared by all candidates; black arrows are the two varying directions. The prediction target vis_2 is not a node of this six-node candidate family. The measured green response retains G2 and G3; the measured pol_1 response then retains G3.

Table S3. Relationship between published methods and evaluated components

Related method	Component evaluated here	Scope
ABCI / Toth et al. (2022)	Graph-information objective within the common candidate model; Sections 5.2--5.4	Objective comparison; full ABCI pipeline not implemented
ABCD / Agrawal et al. (2019)	Information about an author-chosen attribution sign pattern; Sections 5.2--5.3	Adapted objective; full ABCD pipeline not implemented
MEC Shapley / Roos et al. (2026)	Candidate vectors computed with the same underlying evaluator; Section 5.7	Internal representation check; no independent estimator benchmark
Causal SHAP / Ng et al. (2025)	Local linear-Gaussian causal-strength weighting and normalization; Section 5.7	Adaptation with a different attribution target

These comparisons evaluate specified objectives and planning rules under a common finite candidate-model framework. They do not constitute an end-to-end benchmark of complete ABCI, ABCD, or MEC Shapley implementations.

References

Agrawal, R., Squires, C., Yang, K., Shanmugam, K., and Uhler, C. (2019). ABCD-Strategy: Budgeted Experimental Design for Targeted Causal Structure Discovery. AISTATS, PMLR 89.

Holm, S. (1979). A Simple Sequentially Rejective Multiple Test Procedure. Scandinavian Journal of Statistics, 6, 65--70.

Ng, W. Y., Wang, L. R., Liu, S., and Fan, X. (2025). Causal SHAP: Feature Attribution with Dependency Awareness through Causal Discovery. IJCNN.

Roos, D., Gerwinn, S., van de Meent, J.-W., and Magliacane, S. (2026). Improving Local Explainability By Learning Causal Graphs From Data. TMLR.

Toth, C., Lorch, L., Knoll, C., Krause, A., Pernkopf, F., Peharz, R., and von Kügelgen, J. (2022). Active Bayesian Causal Inference. NeurIPS 35.

Wilcoxon, F. (1945). Individual Comparisons by Ranking Methods. Biometrics Bulletin, 1(6), 80--83.